

[Aim: 100/100 in Maths]



TRIGONOMETRY WORKSHEET DISCUSSION CLASS-10TH



1. If $4 \tan \theta = 5 \sin \theta$ where $0^\circ < \theta < 90^\circ$, then the value of $\sin^2 \theta - \cos^2 \theta$ is

(a) $\frac{7}{25}$

(b) $-\frac{7}{25}$

(c) $\frac{1}{25}$

(d) $\frac{16}{25}$

$$4 \tan \theta = 5 \sin \theta$$

$$4 \frac{\cancel{s}}{c} = 5 \cancel{s}$$

$$c = \frac{4}{5}$$

$$s = \frac{3}{5}$$

2. For an acute angle θ , which of the following is not possible?

(a) $\sin\theta = \frac{3}{5}$

(b) $\sec\theta = \frac{1}{2}$

(c) $\tan\theta = 5$

(d) $\cot\theta = \sqrt{3}$

3. If $\sin\theta = \frac{a}{b}$, where $0 < a < b$, then $\sec\theta + \tan\theta$ is equal to

(a) $\sqrt{\frac{b+a}{b-a}}$

(b) $\sqrt{\frac{b-a}{b+a}}$

(c) $\frac{a+b}{b-a}$

(d) $\frac{b}{b-a}$

4. **Assertion (A)** : For every acute angle θ $\sin\theta + \cos\theta \leq \sqrt{2}$.

Reason (R) : $(\sin\theta + \cos\theta)^2 = 1 + 2\sin\theta \cos\theta$, and $2\sin\theta \cos\theta \leq \sin^2\theta + \cos^2\theta$ for all θ .

- (a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

$$(\sin\theta + \cos\theta)^2 = \sin^2\theta + \cos^2\theta + 2\sin\theta\cos\theta$$

$$(\sin\theta + \cos\theta)^2 = 1 + 2\sin\theta\cos\theta \quad \checkmark - (1)$$

we know

$$(\sin\theta - \cos\theta)^2 \geq 0$$

$$\sin^2\theta + \cos^2\theta - 2\sin\theta\cos\theta \geq 0$$

$$\sin^2\theta + \cos^2\theta \geq 2\sin\theta\cos\theta$$

$$2\sin\theta\cos\theta \leq \sin^2\theta + \cos^2\theta$$

$$2\sin\theta\cos\theta \leq 1$$

$$\sin\theta + \cos\theta = \sqrt{1 + 2\sin\theta\cos\theta}$$

$$\leq \sqrt{2} \leq 2$$

5. If $\tan\theta = 24$, find the value of $\frac{\sin\theta + \cos\theta}{\sin\theta - \cos\theta}$

6. If $3\sin^2\theta + 7\cos^2\theta = 4$ and θ is acute, find θ . Hence find the value of $\tan \theta + \cot \theta$.

$$(3+4)$$

$$\underline{3\sin^2\theta + 3\cos^2\theta + 4\cos^2\theta = 4}$$

$$3(\sin^2\theta + \cos^2\theta) + 4\cos^2\theta = 4$$

$$3 + 4\cos^2\theta = 4$$

$$4\cos^2\theta = 1$$

$$\cos^2\theta = \frac{1}{4}$$

$$\cos\theta = \frac{1}{2} \Rightarrow \theta = 60^\circ$$

7. Evaluate : $\frac{2\sin^2 60^\circ + 3\tan^2 30^\circ - \operatorname{cosec}^2 45^\circ}{\cot^2 30^\circ + \sec^2 60^\circ - \operatorname{cosec}^2 30^\circ}$

8. If $\sin(A - B) = \frac{1}{2}$ and $\cos(A + B) = \frac{1}{2}$. Where $0^\circ < A + B \leq 90^\circ$ and $A > B$, find A and B.

✓

9. It is given that $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$ and $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$. Using these, evaluate $\frac{3\sin 10^\circ - 4\sin^3 10^\circ}{4\cos^3 10^\circ - 3\cos 10^\circ}$.

$$\Rightarrow \frac{\sin 3(10)}{\cos 3(10)}$$

$$\Rightarrow \frac{\sin 30}{\cos 30}$$

$$\tan 30$$

$$\rightarrow \frac{1}{\sqrt{3}}$$

$$\rightarrow \frac{\frac{1}{2}}{\frac{\sqrt{3}}{2}}$$

$$\rightarrow \frac{1}{\sqrt{3}}$$

10. If $\sin^4\theta + \cos^4\theta = \frac{1}{2}$ and θ is acute, find the value of θ .

$$\underbrace{(\sin^2\theta)}_A + \underbrace{(\cos^2\theta)}_B = \frac{1}{2}$$
$$A^2 + B^2$$

$$\Rightarrow (A+B)^2 - 2AB$$

$$\Rightarrow (\sin^2\theta + \cos^2\theta)^2 - 2\sin^2\theta\cos^2\theta = \frac{1}{2}$$

$$\Rightarrow \underbrace{(1)^2 - 2\sin^2\theta\cos^2\theta} = \frac{1}{2}$$

$$1 - \frac{1}{2} = 2\sin^2\theta\cos^2\theta$$

$$\frac{1}{2} = 2\sin^2\theta\cos^2\theta$$
$$\frac{1}{4} = (\sin\theta\cos\theta)^2$$
$$\sqrt{\frac{1}{4}} = \sin\theta\cos\theta$$

$$\boxed{\sin\theta\cos\theta = \frac{1}{2}}$$

we know,

$$(\sin \theta - \cos \theta)^2 = \sin^2 \theta + \cos^2 \theta - 2 \sin \theta \cos \theta$$

$$\begin{aligned} (\sin \theta - \cos \theta)^2 &= 1 - \cancel{2} \left(\frac{1}{\cancel{2}} \right) \\ &= 1 - 1 \end{aligned}$$

$$\sin \theta - \cos \theta = 0$$

$$\sin \theta = \cos \theta$$

$$\frac{\sin \theta}{\cos \theta} = 1$$

$$\boxed{\tan \theta = 1} \Rightarrow \boxed{\theta = 45^\circ}$$

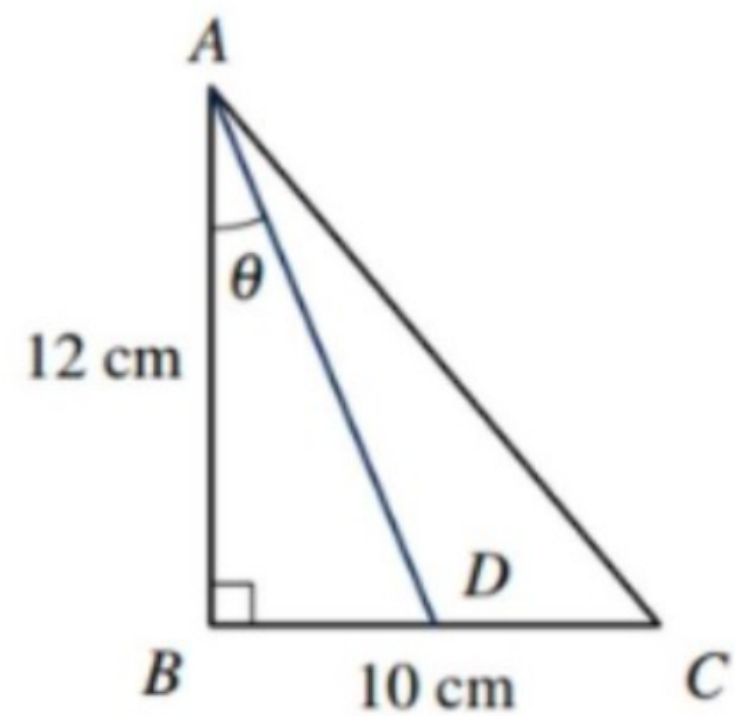
11. In $\triangle ABC$, right-angled at B, $AB = 12$ cm and $BC = 10$ cm. D is the mid-point of BC and $\angle BAD = \theta$. Find:

(i) $\sin \theta$

(ii) $\cos \theta$

(iii) $\tan \theta + \cot \theta$

(iv) $\cot^2 \theta - \operatorname{cosec}^2 \theta$



Handwritten mark resembling a stylized 'd' or a checkmark.

12. Prove that $\frac{1}{1-\tan^2 A} + \frac{1}{1-\cot^2 A} = 1$, where $A \neq 45^\circ$.

$\frac{s}{c}$

$\frac{c}{s}$

✓

13. Prove that $\frac{\sin \theta}{1 - \cot \theta} + \frac{\cos \theta}{1 - \tan \theta} = \sin \theta + \cos \theta$.

$\frac{\sin \theta}{1 - \cot \theta}$

$\frac{\cos \theta}{1 - \tan \theta}$

14. Prove that $\sqrt{\frac{1-\sin \theta}{1+\sin \theta}} + \sqrt{\frac{1+\sin \theta}{1-\sin \theta}} = 2\sec \theta.$

✓

15. Prove that $(\sin A \cos B + \cos A \sin B)^2 + (\cos A \cos B - \sin A \sin B)^2 = 1$.

LHS

$$(\sin A \cos B + \cos A \sin B)^2 + (\cos A \cos B - \sin A \sin B)^2$$

$$\Rightarrow \underbrace{\sin^2 A \cos^2 B} + \underbrace{\cos^2 A \sin^2 B} + \cancel{2 \sin A \cos B \cos A \sin B} + \underbrace{\cos^2 A \cos^2 B} + \underbrace{\sin^2 A \sin^2 B} - \cancel{2 \cos A \cos B \sin A \sin B}$$

$$\Rightarrow \sin^2 A (\underbrace{\cos^2 B + \sin^2 B}_{=1}) + \cos^2 A (\underbrace{\cos^2 B + \sin^2 B}_{=1})$$

$$\Rightarrow \sin^2 A + \cos^2 A$$

$$\Rightarrow \textcircled{1} \textcircled{HP}$$

16. If $x = a \sec \theta + b \tan \theta$ and $y = a \tan \theta + b \sec \theta$, prove that $x^2 - y^2 = a^2 - b^2$.

17. If $\sec \theta + \tan \theta = l$, prove that $\sec \theta = \frac{l^2+1}{2l}$. Hence show that $\sin \theta = \frac{l^2-1}{l^2+1}$.

$$\textcircled{+} \quad \sec \theta - \tan \theta = \frac{1}{l}$$

$$2 \sec \theta = l + \frac{1}{l}$$

$$2 \sec \theta = \frac{l^2+1}{l}$$

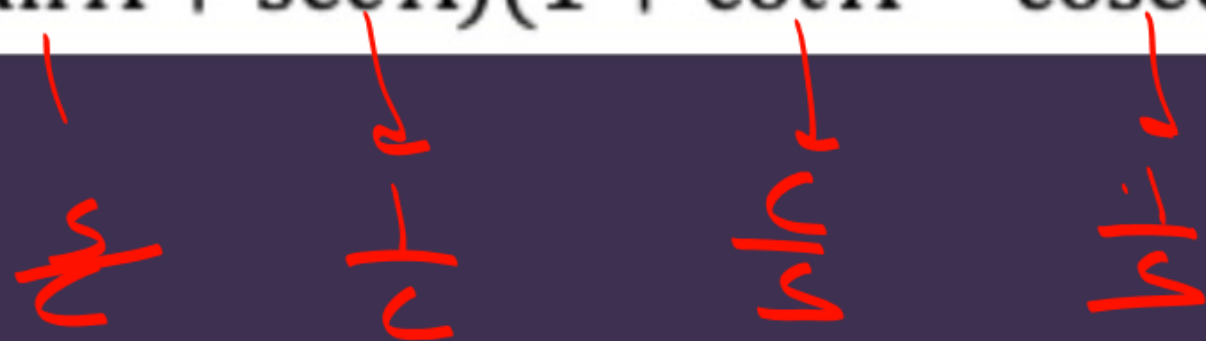
$$\sec \theta = \frac{l^2+1}{2l}$$

$$\sin \theta$$

18. Prove that : $\frac{\cot \theta + \operatorname{cosec} \theta - 1}{\cot \theta - \operatorname{cosec} \theta + 1} = \frac{1 + \cos \theta}{\sin \theta}$.

Handwritten red note: $\rightarrow \operatorname{cosec}^2 \theta - \cot^2 \theta$

19. Prove that $(1 + \tan A + \sec A)(1 + \cot A - \operatorname{cosec} A) = 2$.



20. Prove that : $\frac{\tan^3 \theta}{1 + \tan^2 \theta} + \frac{\cot^3 \theta}{1 + \cot^2 \theta} = \sec \theta \operatorname{cosec} \theta - 2 \sin \theta \cos \theta$.

LHS of s, c

$$\begin{aligned} & \Rightarrow \frac{\frac{\sin^3}{\cos^3}}{1 + \frac{\sin^2}{\cos^2}} + \frac{\frac{\cos^3}{\sin^3}}{1 + \frac{\cos^2}{\sin^2}} \\ & \Rightarrow \frac{\frac{\sin^3}{\cos^3}}{\frac{\cos^2 + \sin^2}{\cos^2}} + \frac{\frac{\cos^3}{\sin^3}}{\frac{\sin^2 + \cos^2}{\sin^2}} \\ & \Rightarrow \frac{\sin^3}{\cos} \cdot \frac{\cos^2}{\cos^2 + \sin^2} + \frac{\cos^3}{\sin} \cdot \frac{\sin^2}{\sin^2 + \cos^2} \end{aligned}$$

$$\begin{aligned} & \Rightarrow \frac{\sin^3 \cdot \cancel{\cos^2}}{\cancel{\cos^2} + \sin^2} + \frac{\cancel{\cos^3} \cdot \sin^2}{\sin^2 + \cancel{\cos^2}} \\ & \Rightarrow \frac{\sin^3}{\cos} + \frac{\cos^3}{\sin} \\ & \Rightarrow \frac{\sin^4 + \cos^4}{\sin \cdot \cos} \\ & \Rightarrow \frac{1 - 2\sin^2 \cos^2}{\sin \cdot \cos} \\ & \Rightarrow \frac{1}{\sin \cdot \cos} - \frac{2\sin^2 \cos^2}{\sin \cdot \cos} \\ & \Rightarrow \sec \theta \operatorname{cosec} \theta - 2 \sin \theta \cos \theta \quad \text{Proved} \end{aligned}$$

21. Prove that $\frac{\sin^4 \theta + \cos^4 \theta - 1}{\sin^6 \theta + \cos^6 \theta - 1} = \frac{2}{3}$.

Num $\Rightarrow$

~~$-2\sin^2 \theta \cos^2 \theta$~~
 $-2\sin^2 \theta \cos^2 \theta$

D $\rightarrow$

$\underbrace{(\sin^2 \theta)^3}_{a^3} + \underbrace{(\cos^2 \theta)^3}_{b^3} - 1$

$(a+b)^3 - 3ab(a+b) - 1$

$\Rightarrow (\sin^2 \theta + \cos^2 \theta)^3 - 3\sin^2 \theta \cos^2 \theta (\sin^2 \theta + \cos^2 \theta) - 1$

~~$-3\sin^4 \cos^2 \theta(1)$~~

$\boxed{D \Rightarrow -3\sin^2 \theta \cos^2 \theta}$

$\frac{N}{D} \Rightarrow \frac{-2\sin^2 \theta \cos^2 \theta}{-3\sin^2 \theta \cos^2 \theta}$

$\Rightarrow \frac{2}{3} = \text{RHS}$

HP

$$(a+b)^3 = \underline{a^3 + b^3} + \underline{3ab(a+b)}$$

$$(a+b)^3 - 3ab(a+b) = a^3 + b^3$$

22. If $x \sin^3 \theta + y \cos^3 \theta = \sin \theta \cos \theta$ and $x \sin \theta = y \cos \theta$, prove that $x^2 + y^2 = 1$.

$$\Rightarrow \underline{x} \underline{s}^3 + \left(\underline{x} \underline{\frac{s}{c}} \right) c^3 = s \cdot c$$

$$\Rightarrow x [s^2 + c^2] = s \cdot c$$

$$\Rightarrow x(1) = c$$

$$\boxed{x = \cos \theta}$$

$$\boxed{x \frac{s}{c} = y}$$

$$y = \cos \theta \frac{\sin \theta}{\cos \theta}$$

$$\boxed{y = \sin \theta}$$

LHS

$$c^2 + s^2$$

$$\Rightarrow 1 = \text{RHS}$$

HP

23. ~~Q~~
 No. 15.

If $\csc \theta - \sin \theta = m$ and $\sec \theta - \cos \theta = n$, prove that $(m^2 n)^{2/3} + (mn^2)^{2/3} = 1$.

$$m = \frac{1}{s} - s$$

$$m = \frac{1-s^2}{s}$$

$$m = \frac{c^2}{s}$$

$$n = \frac{1}{c} - c$$

$$n = \frac{1-c^2}{c}$$

$$n = \frac{s^2}{c}$$

$$c^2 + s^2$$

$$\Rightarrow 1 = \text{RHS} \quad \text{HP}$$

$(\frac{m}{n})^3$
 $\rightarrow \frac{m^3}{n^3}$

$$(m^2 n)^{2/3} \Rightarrow \left[\left(\frac{c^2}{s} \right)^2 \cdot \frac{s^2}{c} \right]^{2/3}$$

$$\Rightarrow \left[\frac{c^4}{s^2} \cdot \frac{s^2}{c} \right]^{2/3} \Rightarrow (c^3)^{2/3} \Rightarrow c^2$$

$$(mn^2)^{2/3} \Rightarrow \left[\frac{c^2}{s} \cdot \left(\frac{s^2}{c} \right)^2 \right]^{2/3}$$

$$\Rightarrow \left[\frac{c^2}{s} \cdot \frac{s^4}{c^2} \right]^{2/3}$$

$$\Rightarrow (s^3)^{2/3} \Rightarrow s^2$$

24. If $\operatorname{cosec} \theta = x + \frac{1}{4x}$, prove that $\operatorname{cosec} \theta + \cot \theta = 2x$ or $\frac{1}{2x}$.

$$\operatorname{cosec}^2 \theta - \cot^2 \theta = 1$$

$$\operatorname{cosec}^2 \theta - 1 = \cot^2 \theta$$

$$\Rightarrow \left(x + \frac{1}{4x}\right)^2 - 1 = \cot^2 \theta$$

$$\Rightarrow x^2 + \frac{1}{16x^2} + 2x \left(\frac{1}{4x}\right) - 1 = \cot^2 \theta$$

$$\Rightarrow x^2 + \frac{1}{16x^2} - \frac{1}{2} = \cot^2 \theta$$

$$\Rightarrow \underbrace{\left(x\right)^2 + \left(\frac{1}{4x}\right)^2 - 2\left(x\right)\left(\frac{1}{4x}\right)}_{a^2 + b^2 - 2ab} = \cot^2 \theta$$

$$\left(x - \frac{1}{4x}\right)^2 = \cot^2 \theta$$

$$\cot \theta = \pm \left(x - \frac{1}{4x}\right)$$

$$\cot \theta = + \left(x - \frac{1}{4x}\right)$$

$$\cot \theta = - \left(x - \frac{1}{4x}\right)$$

$$\operatorname{cosec} \theta + \cot \theta$$

$$\Rightarrow x + \frac{1}{4x} + x - \frac{1}{4x} = 2x$$

$$\operatorname{cosec} \theta + \cot \theta$$

$$\Rightarrow x + \frac{1}{4x} - x + \frac{1}{4x} = \frac{1}{2x}$$

Q55

If $\sin \theta + \sin^2 \theta = 1$, find the value of $\cos^{12} \theta + 3\cos^{10} \theta + 3\cos^8 \theta + \cos^6 \theta - 1$.

$$\sin \theta + \sin^2 \theta = \sin^2 \theta + \cos^2 \theta \Rightarrow$$

$$\sin \theta = \cos^2 \theta$$

$$\cos^2 \theta = t$$

$$t^6 + 3t^5 + 3t^4 + t^3 - 1$$

$$\Rightarrow t^3 [t^3 + 3t^2 + 3t + 1] - 1$$

$$\Rightarrow t^3 (t+1)^3 - 1$$

$$\Rightarrow [t(t+1)]^3 - 1$$

$$\Rightarrow [\cos^2 \theta (\cos^2 \theta + 1)]^3 - 1$$

$$\Rightarrow [\sin \theta (\sin \theta + 1)]^3 - 1$$

$$+ (\sin^2 \theta + \sin \theta)^3 - 1$$

$$(a+b)^3 = a^3 + b^3 + 3a^2b + 3ab^2$$

$\Rightarrow 0$

HW → Worksheet

V. mp

THANK YOU COODIESS !!

